Name _____

CSCI 332, Fall 2026
Quiz 1

1. (2 points) For each symbol below, write its name or meaning in English. The first is filled in for you as an example.

   (a) $\mathscr{P}$: power set

   (b) $\mathbb{R}$: _____

   (c) $\cap$: _____

   (d) $\cup$: _____

   (e) $|\ |$: _____

2. (3 points) Circle True or False for each of the following statements.

   (a) $a \in \{\{a\}, b\}$: True or False

   (b) $\{1,2\} \subseteq \{1,2,3,5\}$: True or False

   (c) $11 \in \{x \in \mathbb{Z} : x < 10\}$: True or False

3. (5 points) Given the following preference lists as input for the hospital/medical student matching problem.

	1st choice	2nd choice	3rd choice	4th choice
H1	S4	S1	S2	S3
H2	S4	S2	S3	S1
H3	S3	S2	S1	S4
H4	S3	S2	S1	S4
	1st choice	2nd choice	3rd choice	4th choice
S1	H3	H4	H2	H1
S2	H4	H1	H2	H3
S3	H1	H2	H3	H4
S4	H3	H4	H2	H1

   (a) Is the matching $M = \{(H1, S3), (H2, S4), (H3, S1), (H4, S2)\}$ stable? Circle one: Yes or No

   (b) How do you know?